

	Phenomic layer	Genomic layer	Interactome layer
NATIVE DATA STRUCTURE	 Spatial images with local, repeated visual patterns	 Ordered marker dosages; signal is largely additive	 Large relational, hierarchical networks can exceed one context window
ARCHITECTURAL BIAS	CNN / U-Net locality + shared filters	Transformer order + contextual attention	Language model bounded sequential context
DEFAULT ALIGNMENT	STRONG  Spatial assumptions hold	PARTIAL  Order holds; language-like context is limited	WEAK DIRECT FIT  Graph structure is not sequential
REQUIRED ADAPTATION	Task-specific tiling and scarce-label segmentation	Joint marker–environment tokens encode $G \times E$	Graph-based pretraining + tool-based exploration
DISCOVERY TASK	MEASUREMENT Chapter 2 · pennycress	PREDICTION Chapter 3 · $G \times E$	INTERPRETATION Chapter 4 · MENTOR-RL